

Project Documentation: Business Sales Performance Dashboard

1. Executive Summary

This project involves the development of an interactive Business Sales Dashboard designed to provide a 360-degree view of retail performance. By leveraging **SQL** for data transformation and **Power BI** for visualization, this dashboard empowers stakeholders to track revenue trends, customer demographics, and product performance across multiple regions.

2. Key Performance Indicators (KPIs)

The dashboard highlights six primary metrics to assess business health at a glance:

- **Total Revenue:** \$14.16M
- **Total Orders:** 186
- **Average Order Value (AOV):** \$76.15K
- **Gender Distribution:** 61.01% Female vs. 38.99% Male
- **Top Year:** 2023

3. Deep-Dive Insights

A. Sales Seasonality (Month-Wise)

- **Peak Performance:** September emerged as the highest-grossing month (\$1,648K), followed closely by February and July.
- **Observation:** There is a notable dip in March (\$797K) and November (\$728K). This suggests a need for targeted marketing campaigns or "Pre-Holiday" sales during these shoulder months to stabilize cash flow.

B. Regional Performance (City-Wise)

- **Market Leader:** Quetta leads the sales charts with \$3,656K.
- **Geographic Spread:** Sales are relatively well-distributed across major hubs including Islamabad, Lahore, Peshawar, and Karachi. Quetta and Islamabad combined contribute significantly to the total revenue.

C. Product & Payment Analysis

- **Top Products:** Peripherals drive the highest volume, with **Mouses** (\$2,729K),
- **Monitors** (\$2,254K), and **Printers** (\$2,414K) being the top three revenue generators.

- **Payment Preferences:** The customer base shows a strong preference for digital transactions, with **Card** (37.5%) and **Online** (34.6%) payments making up over 70% of the total volume.

D. Customer Demographics

- **Age Range:** The "Under 60" and "Under 50" age brackets are the primary revenue drivers, accounting for over \$7.3M combined.
- **Audience:** The 61% female customer base suggests that marketing creative and product positioning should remain aligned with this dominant demographic.

4. Technical Stack

- **SQL Server:** Used for data extraction, cleaning (handling nulls, formatting dates), and aggregating tables to optimize Power BI performance.
- **Power BI DAX:** Implemented measures for AOV, Month-over-Month growth, and dynamic filtering.
 - **Data Modeling:** Established relationships between Sales, Product, and Customer tables.
 - **UI/UX:** Designed a high-contrast dark mode interface for better readability and modern aesthetic.

5. Recommendations

1. **Inventory Optimization:** Increase stock levels for top-performing electronics (Mouses, Printers) before the Q3 peak in September.
2. **Loyalty Program:** Target the "Under 60" demographic with a loyalty program, as they represent the highest lifetime value.
3. **Regional Expansion:** Investigate the specific success factors in Quetta to replicate those strategies in lower-performing regions like Karachi.